TripShare

Software Requirement Specifications

Bachelor of Science in Software Engineering

By

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document will lay out the software requirements for the device named TripShare – Android Based Trip Management Application (Version 1.0). This document explains the functional and non-functional requirements of TripShare in detail.

This SRS is an all-inclusive document containing the design of the whole TripShareMobile application from user authentication, trip creation and managing, expense monitoring and sharing costs, plan itinerary and changing route in normal mode to take photos/videos during a trip as well as sharing them, use blockchain for effectiveness of data to generate reports. The paper is intended as a handbook for system planning and designing, developing, testing and validating the TripShare application.

1.2 Document Conventions

This SRS document adheres to the IEEE Software Requirements Specification standard. The following are the rules of the documentation:

- Section headings and subheadings are numbered hierarchically for clarity and easy reference.
- Functional requirements are labeled as FR1, FR2, FR3, and so on.
- Performance, security, usability, scalability, and reliability are categorized as non-functional requirements.
- The keyword “shall” is used to indicate mandatory system requirements.
- All requirements are written in clear, concise, and unambiguous language.
- Priorities of high-level requirements apply to the detailed requirements, related to them unless stated otherwise.

1.3 Intended Audience and Reading Suggestions

The following are the intended audiences for this document:

- **Project Supervisor:** To review and approve the sysTem requirements.
- **Developers:** To know the system functionality and constraints for implementation.
- **Testers:** to verify and validate system behavior against specified requirements.
- **Examiners and Evaluators:** Assess the completeness and truth of the SRS document.

1.4 Product Scope

TripShare is an Android-based mobile application developed to simplify and enhance group trip management. The application consolidates multiple travel-related features into a single platform, allowing users to efficiently plan trips, manage and split shared expenses, and navigate routes using Google Maps. In addition, TripShare enables users to capture, store, and organize trip photos and videos, ensuring that all travel details and memories are conveniently accessible in one place.

The most important feature of TripShare is the advocacy of authentic and verified tourism by applying blockchain technology to validate and secure user-created content about their trips. The app has also

a Public Trip Gallery where one can share confirmed photos and videos with the givers of the travel community.

Above all, TripShare is aimed at deploying a secure, user-friendly, and scalable solution that will reduce the usage of various travel apps and increase the collaboration between travelers and the digital tourism strategies in the contemporary environment.

2. Overall Description

2.1 Product Perspective

TripShare is both a recently created Android based mobile application that was built under a Final Year Project to solve the typical group travel challenges. The application will be developed as a built in travel management system where trip planning, expenses management, route navigation, media sharing, and content authenticity checking will be combined on a single platform.

Today, the passengers rely on various standalone apps such as splitters of expenses, navigation applications, and social media in an attempt to control various aspects of the journeys. To this effect, TripShare is not competing with an existing system but rather the connection between the fragmented tools can be fulfilled with an all-encompassing solution. Integration reduces complexity and is more useful and collaborative with travelers.

At the system architectural level, TripShare is an independent mobile application which communicates with various services. These are Firebase Authentication to enable a safe logon and a user account management system, Firebase Firestore and Cloud storage to ensure data is saved, and Google Maps API to provide real-time navigation and location services. Besides that, a hashing scheme that is based on blockchain is also included to safeguard the authenticity and integrity of user-created materials, including photos, videos, and travel review.

Primary TripShare application serves as a core controller, which maintains the interaction with the user, business logic, and data flow among core modules and external services. The system is also independent and is more or less a modular application although much of its functionality is provided via third-party APIs, meaning it does not depend on them to create a trip, calculate expenses, or manage media, making it somewhat independent.

2.2 Product Functions

TripShare has a variety of features which, when combined, will facilitate full trip management. On the system-wide level, the key functions of the system are the following:

- **User Registration and Authentication:** It is an option that enables the creation of user accounts and secure authentication by Firebase Authentication. This is to make sure that data about trips and shared resources are seen by the authorized users.
- **Trip Management-Creation:** This feature lets the user make new trips, indicating the destination, the date, budget and description. The organizers are also in a position to update the information of their trips and handle the group members at the same time.

- **Team Collaboration and Invitations:** The user may offer their friends or family members unique invitations by inviting them to a trip. All the members of the trip will see the information provided as they will have access to it.
- **Expense Tracking and Cost Splitting:** Records recorded expenses of food, accommodation, fuel and ticket expenses. The system is used to compute individual shares automatically and show pending balances, over payments or settlements.
- **Route Planning and Navigation:** This uses Google Maps API in displaying routes, places, tourist attraction sites, restaurants, and hotels/hostels. It aids the customers to map out effective traveling routes.
- **Media Capture and Storage:** This feature allows users to capture, upload, and store trip photos and videos directly within the application. All media files are securely stored using cloud storage to ensure data safety and accessibility.
- **Blockchain-Based Content Verification:** The system generates cryptographic hashes for uploaded media and user reviews, ensuring content authenticity and protecting against tampering or unauthorized modifications.
- **Public Trip Gallery:** This module displays verified photos and videos shared by travelers, enabling users to explore destinations through genuine travel experiences rather than manipulated or misleading content.
- **Report Generation:** The application provides downloadable trip summaries and expense reports in both PDF and CSV formats. These reports can be easily shared via email or messaging applications.

2.3 User Classes and Characteristics

TripShare is designed to support multiple user classes, each with distinct roles, usage patterns, and levels of technical expertise.

2.3.1 General Travelers

General travelers are occasional users who utilize the application to plan trips, track and manage shared expenses, and store memories from their journeys. Most users in this category possess basic smartphone proficiency; therefore, the application emphasizes a simple, intuitive, and user-friendly interface. These users prioritize ease of use and quick access to essential features over advanced customization options.

2.3.2 Group Trip Members

The members of the group travel are actively involved in common traveling by contributing costs, sharing routes, publishing media, and engaging fellow group members. This application is highly utilized by such users during a trip, and real-time updates are not only trusted, but also heavily used as well. They must have frequent synchronization and precision on the calculation of expenses.

2.3.3 Trip Organizers

The organizers of the trips have the responsibility of creating trips, inviting, and detailing and monitoring of the expenses of trips. They require administrative capabilities: editing of trip data, permission management and reporting. It is a type of user key to the functionality of the system.

2.3.4 Travel Vloggers and Content Contributors

Users in this category are primarily interested in publishing travel photos, videos, and recommendations to the public community. They place high importance on content visibility, credibility, and authenticity. For these users, blockchain-based content verification plays a crucial role in establishing trust and preventing content manipulation. Among all user classes, group trip members and trip organizers represent the primary users of TripShare, as the majority of functional requirements are designed to support collaborative trip planning, expense management, and shared media handling.

2.4 Operating Environment

The TripShare application is designed to work in this type of environment:

- **Hardware Platform:** Android smartphones and tablets equipped with GPS, camera, and internet connectivity.
- **Operating System:** Android OS version 8.0 (Oreo) or newer.
- **Development Environment:** Android Studio IDE.
- **Backend Infrastructure:** Firebase Authentication, Firebase Firestore/Realtime Database, Firebase Cloud Storage.
- **Third-party Services:** Google Maps API for navigation and location-based features.
- **Network Environment:** Wi-Fi or mobile data connection The App is optimized for mobile devices, supporting multiple screen sizes to ensure maximum compatibility across a wide range of Android devices.

2.5 Design and Implementation Constraints

Following are several constraints on the design and development of TripShare:

- The application should be developed solely for Android.
- Cloud sync, navigation, and blockchain verification require constant internet connectivity.
- Usage of Google Maps API and Firebase services will be under respective usage limits and policies.
- Implementation of blockchain must be lightweight to ensure maximum performance on mobile devices.
- The system must be designed to meet the standards for security set by Android and regulations concerning data privacy.
- UI design should be consistent with Android Material Design for better usability. These constraints influence architectural decisions and limit certain implementation choices.

2.6 User Documentation

Following is the user documentation to be provided with the TripShare application:

- Comprehensive User Manual (PDF).
- In-app contextual help and tooltips.
- Installation and configuration guide.

User Documentation and Guidance: The system shall provide step-by-step guidance for core functionalities, including trip creation and expense tracking. The documentation shall be designed to support users with different levels of technical expertise.

2.7 Assumptions and Dependencies

The following assumptions and dependencies are considered while developing the TripShare system:

- Users have Android devices that are GPS and internet-enabled.
- Google Maps API, Firebase services, and blockchain components are accessible and functional.
- Users grant permissions like camera access, storage access, and location access.
- Changes in third-party service policies might need system updates.
- The cloud infrastructure can be scaled to meet increased demand from its users.

Failure of these assumptions can affect system functionality and performance.

Following is the user documentation to be provided with the TripShare application:

- Installation Instructions (PDF).
- In-app contextual help and tooltips.

Installation and configuration guide • Step-by-step tutorials on core features of the site, including creating a trip and tracking expenses. The documentation will be designed to support users with varying technical backgrounds.

3. External Interface Requirements

3.1 User Interfaces

- **Graphical User Interface (GUI):** The application shall follow Android Material Design principles and support touch-based interactions with consistent use of color schemes, typography, and icons across all screens.
- **Screens:** The system shall provide the following primary screens: Login and Registration, Dashboard, Trip Management, Expense Tracking, Maps and Navigation, Media Gallery, and Public Trip Gallery.
- **User Feedback:** The application shall display clear error messages, input validation prompts, and confirmation notifications to ensure a smooth and user-friendly experience.

3.2 Hardware Interfaces

- **Devices:** Android phones/tablets with GPS and camera.

- **Interaction:** GPS for location, camera for media, local storage for caching.
- **Communication:** Via standard Android APIs, following security permissions.

3.3 Software Interfaces

- **OS:** Android 8.0+
- **Cloud Services:** Firebase Auth, Firestore/Realtime DB, Cloud Storage.
- **APIs:** Google Maps, Blockchain Verification.
- **Data Exchange:** JSON over RESTful services.

3.4 Communications Interfaces

- **Network:** Wi-Fi or mobile data (3G/4G/5G), HTTPS.
- **Data:** Trip information, expenses, media, blockchain hashes.
- **Security:** TLS/SSL encryption, secure tokens, no plaintext credentials.
- **Sync:** Real-time via Firebase; offline caching with automatic synchronization.

4. System Features

4.1 User Authentication

4.1.1 Description and Priority

Login, registration, password reset, and account management procedures.
Priority: High (Benefit: 9, Penalty: 8, Cost: 5, Risk: 4)

4.1.2 Stimulus/Response Sequences

Table 1: S/R Sequence of Anonymous Questioning

User Action	System Response
Opens app	Displays login/registration screen
Enters credentials	Validates via Firebase Authentication
Requests password reset	Sends reset email
Successful login	Redirects to Dashboard
Invalid login	Displays error message

4.1.3 Functional Requirements

Table 2: Functional Req for Anonymous Questioning

ID	Requirement
REQ-1	Users must register using email/password or Google Sign-In
REQ-2	Authenticate users via Firebase Authentication
REQ-3	Password reset triggers secure email links
REQ-4	Invalid login attempts show clear error messages
REQ-5	Sessions expire after 30 minutes of inactivity

4.2 Trip Management

4.2.1 Description & Priority:

Action for creating, modifying, deleting trips, handling participants, viewing trip information.
Priority: High (Benefit: 9, Penalty: 7, Cost: 6, Risk: 5)

4.2.2Stimulus/Response Sequences

Table 3: S/R Sequence of Anonymous Questioning

User Action	System Response
Selects “Create Trip”	Displays trip creation form
Fills trip details	Validates input, stores in Firestore
Edits/deletes trip	Updates/removes trip accordingly
Adds participants	Sends invitation notifications

4.2.3Functional Requirements

Table 4: Functional Req for Anonymous Questioning

ID	Requirement
REQ-6	Create trips with title, description, dates, and location

ID	Requirement
REQ-7	Only trip owner can edit/delete trips
REQ-8	Participants invited via email or in-app notification
REQ-9	Trip details stored in Firestore and available offline
REQ-10	Validate input to prevent invalid or empty fields

4.3 Expense Tracking

4.3.1 Description & Priority:

Manage expenses, compute contributions, and split joint expenses.
Priority: High (Benefit: 9, Penalty: 8, Cost: 5, Risk: 4)

4.3.2 Stimulus/Response Sequences

Table 5: S/R Sequence of Anonymous Questioning

User Action	System Response
Selects “Add Expense”	Displays expense form
Inputs expense	Validates, stores data
Views balances	Calculating individual contributions.
Edits/deletes	Updates totals accordingly

4.3.3Functional Requirements

Table 6: Functional Req for Anonymous Questioning

ID	Requirement
REQ-11	Record expenses with amount, payer, description
REQ-12	Update participant balances in real-time
REQ-13	Prevent negative or invalid expense charges
REQ-14	Store and Retrieve Historical Expenses for Each Trip
REQ-15	Notify participants of new/updated expenses

4.4 Map & Route Planning

4.4.1 Description & Priority:

Offers real-time maps, route information, and points of interest via GPS and Google Maps.
Priority: Medium (Benefit: 8, Penalty: 6, Cost: 5, Risk: 4)

4.4.2Stimulus/Response Sequences

Table 7: S/R Sequence of Anonymous Questioning

User Action	System Response
Opens map	Displays current location and nearby points of interest (POIs)
Sets a destination	Calculates optimal route
Starts navigation	Provides turn-by-turn directions
GPS unavailable	Displays clear error message

4.4.3 Functional Requirements

Table 8: Functional Req for Anonymous Questioning

ID	Requirement
REQ-16	Display user location in real-time on Google Maps
REQ-17	Allow users to search for destinations and attractions
REQ-18	Route planning shall optimize for shortest distance or time
REQ-19	Cache maps offline to allow temporary navigation without network
REQ-20	Display user-friendly alerts for GPS or location errors

4.5 Media Capture & Gallery

4.5.1 Description & Priority:

Allows the taking of photos and videos, the secure storage of these images, and the ability to view them in a private and public album of a journey.

Priority: Medium (Benefit: 7, Penalty: 5, Cost: 5, Risk: 4)

4.5.2 Stimulus/Response Sequences

Table 9: S/R Sequence of Anonymous Questioning

User Action	System Response
Selects “Capture Photo/Video”	Opens camera interface within the app
Captures media	Stores locally and uploads securely to Firebase Storage
Views gallery	Retrieves media for display, sorted by trip
Deletes media	Removes media from local and cloud storage

4.5.3 Functional Requirements

Table 10: Functional Req for Anonymous Questioning

ID	Requirement
REQ-21	Capture photos and videos directly via the app
REQ-22	Upload media securely to Firebase Cloud Storage
REQ-23	Display trip-specific media galleries for users
REQ-24	Public gallery displays verified content only
REQ-25	Display clear error messages for failed capture or upload

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Table 11: Nonfunctional Req for system Performance

Requirement	Rationale
Dashboard/Home screen loads within 2 seconds under normal network conditions	Improves user experience
Trip formation, editing, or deletion is accomplished within 3 seconds	Ensures timely operations
Real-time location updates every 5 seconds	Accurate navigation and trip tracking
Media uploads (≤10 MB) complete within 10 seconds on 4G	Minimizes user waiting time
Handles up to 5,000 concurrent users without affecting performance	Facilitates scalability
Background synchronization will not impair the foreground performance to below 30 frames/sec	Smooth application functioning during actual use

5.2 Safety Requirements

Table 12: Nonfunctional Req for system Safety

Requirement	Rationale
Onlaine posilost aneboffline datovy cach	Prevents data loss
Users cannot edit or delete another user's trips and expenses	Prevents unauthorized actions
Sensitive media content encrypted during storage and transmission	Protects privacy
GPS or wrong routing enters informative notifications	Verifies User Safety
Compliance with GDPR for personal data handling	Legal and regulatory compliance

5.3 Security Requirements

Table 13: Nonfunctional Req for system Security

Requirement	Rationale
User authentication via Firebase Auth required	Ensures only authorized access
All data communication encrypted using HTTPS/TLS	Secures data in transit
No plaintext storage or transmission of sensitive data	Prevents data breaches
Role-based access control for critical operations	Prevents unauthorized activity
Media uploaded to public gallery verified before display	Prevents malicious content sharing
Compliance with GDPR and Android security guidelines	Ensures legal/privacy standards are met

5.4 Software Quality Attributes

Table 14: Nonfunctional Req for system Quality Attributes

Attribute	Requirement / Target
Usability	Very intuitive UI; new customers can easily learn features without training
Reliability	Uptime ≥ 99% monthly; crash rate < 0.5%
Performance	Heavy operations < = 3sec; location updates every 5sec
Maintainability	Modular design; documented APIs for easy updates and bug fixes
Portability	Compatible with Android 8.0+ devices; supports various screen sizes

Attribute	Requirement / Target
Security	Encryption for sensitive data; role-based access for critical actions
Interoperability	Seamless integration with Firebase, Google Maps API, blockchain module
Robustness	Handles network interruptions gracefully; offline caching ensures no data loss
Reusability	Core modules for reuse in future projects

5.5 Business Rules

Table 15: Nonfunctional Req for system Business Rules

Rule	Rationale
Only trip owners can edit/delete trips and manage participants	Maintains control and data integrity
Participants can add expenses/media but cannot delete trips	Enforces access restrictions
Public gallery content must be verified before display	Prevents inappropriate content
Notifications sent only to relevant participants	Avoids spamming users
User authentication required for all trip-related actions	Ensures authorized access
The offline data is automatically synchronized when the connectivity is restored	It maintains consistency and integrity.

5.6 Detailed Subsystem Design

5.6.1 User Authentication Subsystem

Table 16: Functional Overview of the User Authentication

Aspect	Details
Purpose	Handles secure login, registration, password recovery, and session management
Responsibilities	Validate credentials via Firebase Auth; manage sessions; handle password resets; support Google Sign-In
Component Attributes	AuthenticationManager, SessionHandler, FirebaseAuthInterface

Aspect	Details
Behavior	1. User enters credentials → AuthenticationManager validates 2. On success → SessionHandler stores token 3. On failure → display error 4. Password reset triggers Firebase email
Diagrams	Class Diagram: AuthenticationManager, SessionHandler, FirebaseAuthInterface Sequence Diagram: User login flow

5.6.2 Trip Management Subsystem

Table 17: Functional Overview of the Trip Management

Aspect	Details
Purpose	Create, edit, delete trips; manage participants
Responsibilities	Trip CRUD operations; manage invitations; validate trip data; offline caching
Component Attributes	TripManager, ParticipantManager, OfflineCache
Behavior	1. User creates trip → TripManager validates & stores 2. Add participants → ParticipantManager sends notifications 3. Edit/delete → TripManager updates Firestore & OfflineCache
Diagrams	Class Diagram: TripManager, ParticipantManager, OfflineCache Activity Diagram: Trip creation & participant invitation

5.6.3 Expense Tracking Subsystem

Table 18: Functional Overview of the Expense Tracking

Aspect	Details
Purpose	Track trip expenses and balances
Responsibilities	Add/edit/delete expenses; calculate contributions; notify participants
Component Attributes	ExpenseManager, BalanceCalculator, NotificationHandler

Aspect	Details
Behavior	1. Add expense → ExpenseManager stores 2. BalanceCalculator calculates shares 3. NotificationHandler notifies participants
Diagrams	Class Diagram: ExpenseManager, BalanceCalculator, NotificationHandler Sequence Diagram: Adding expense & updating balances

5.6.4 Map & Route Planning Subsystem

Table 19: Functional Overview of the Map & Route Planning

Aspect	Details
Purpose	Provide real-time maps, route planning, and navigation
Responsibilities	Retrieve GPS location; render maps; calculate routes; provide navigation guidance
Component Attributes	MapController, GPSHandler, RoutePlanner
Behavior	1. GPSHandler fetches location → MapController displays 2. User selects destination → RoutePlanner calculates route 3. Navigation started → MapController provides directions
Diagrams	Class Diagram: MapController, GPSHandler, RoutePlanner Activity Diagram: Navigation flow

5.6.5 Media Capture & Gallery Subsystem

Table 20: Functional Overview of the Media Capture & Gallery

Aspect	Details
Purpose	Capture photos/videos and manage galleries
Responsibilities	Capture media; upload to Firebase Storage; manage personal & public galleries

Aspect	Details
Component Attributes	MediaManager, CloudUploader, GalleryController
Behavior	1. Capture media → MediaManager stores locally 2. CloudUploader uploads to Firebase 3. GalleryController displays media 4. Deletion removes from local & cloud storage
Diagrams	Class Diagram:MediaManager, CloudUploader, Sequence Diagram: Capture, Upload, and Display Media In

5.6.6 Blockchain Verification Subsystem

Table 21: Functional Overview of the Blockchain Verification

Aspect	Details
Purpose	Ensure integrity and authenticity of trip data
Responsibilities	Generate cryptographic hashes; verify data integrity; log verification to blockchain
Component Attributes	HashGenerator, VerificationEngine, BlockchainInterface
Behavior	1. Trip/media added → HashGenerator creates hash 2. VerificationEngine validates hash 3. BlockchainInterface logs verification
Diagrams	Class Diagram: HashGenerator, VerificationEngine, BlockchainInterface Sequence Diagram: Data verification flow

6. Diagrams

6.1 Use case Diagram

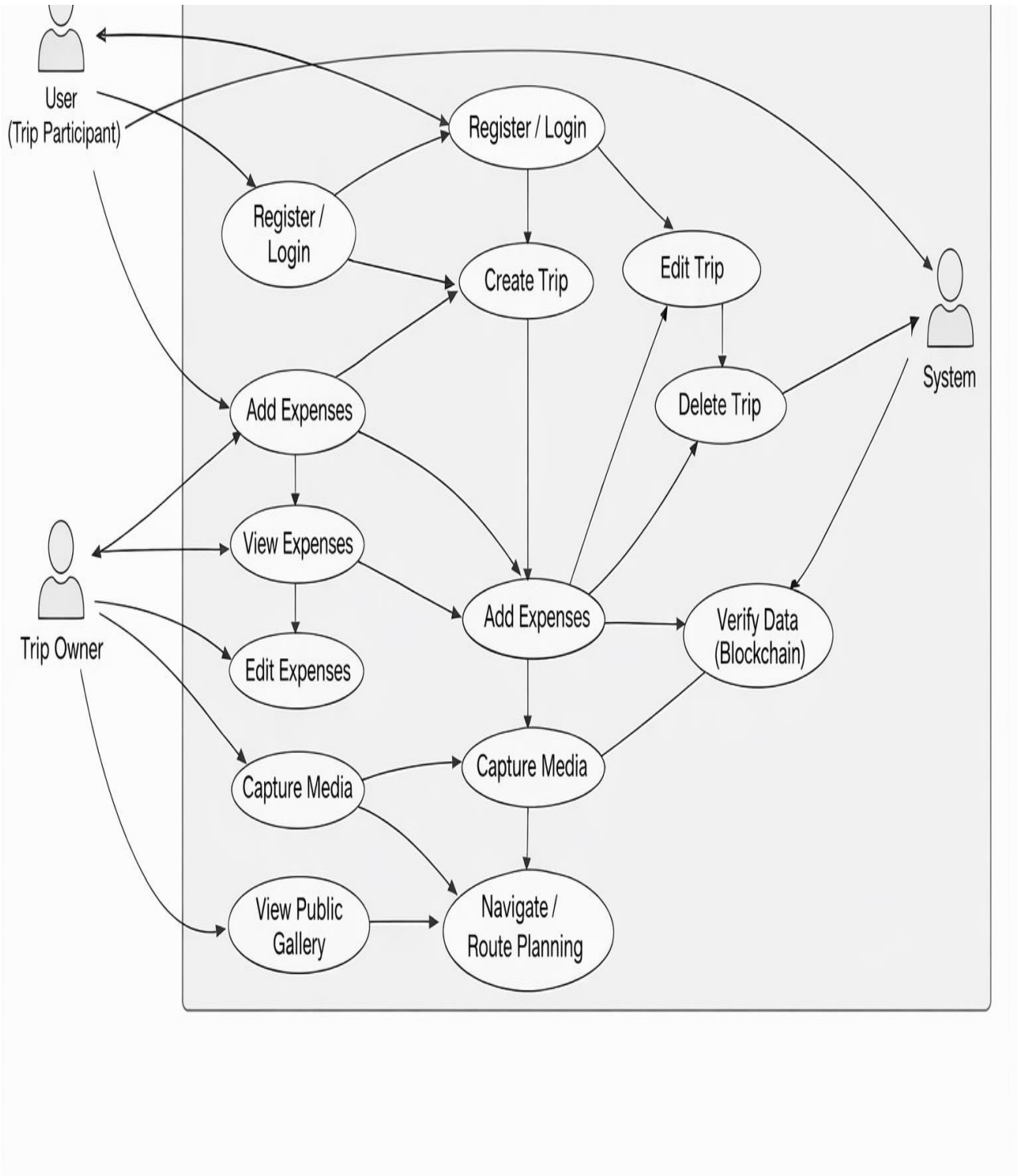

Figure 1: Use Case Diagram showing the System Boundary and Actors Interaction

6.2 ER Diagram

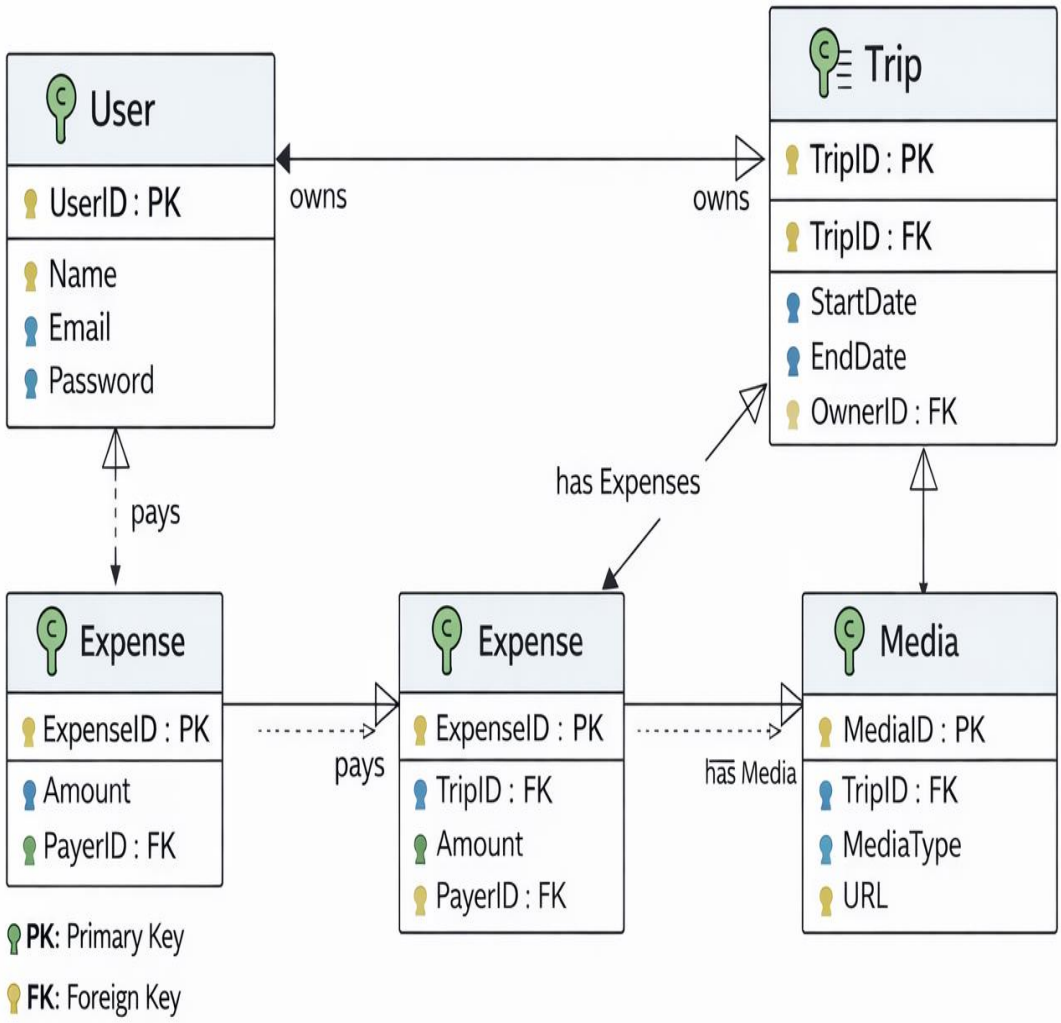

Figure 2: Entity-Relationship Diagram (ERD) showing the Abstract Schematic of the System Data

6.3 Architectural Diagram

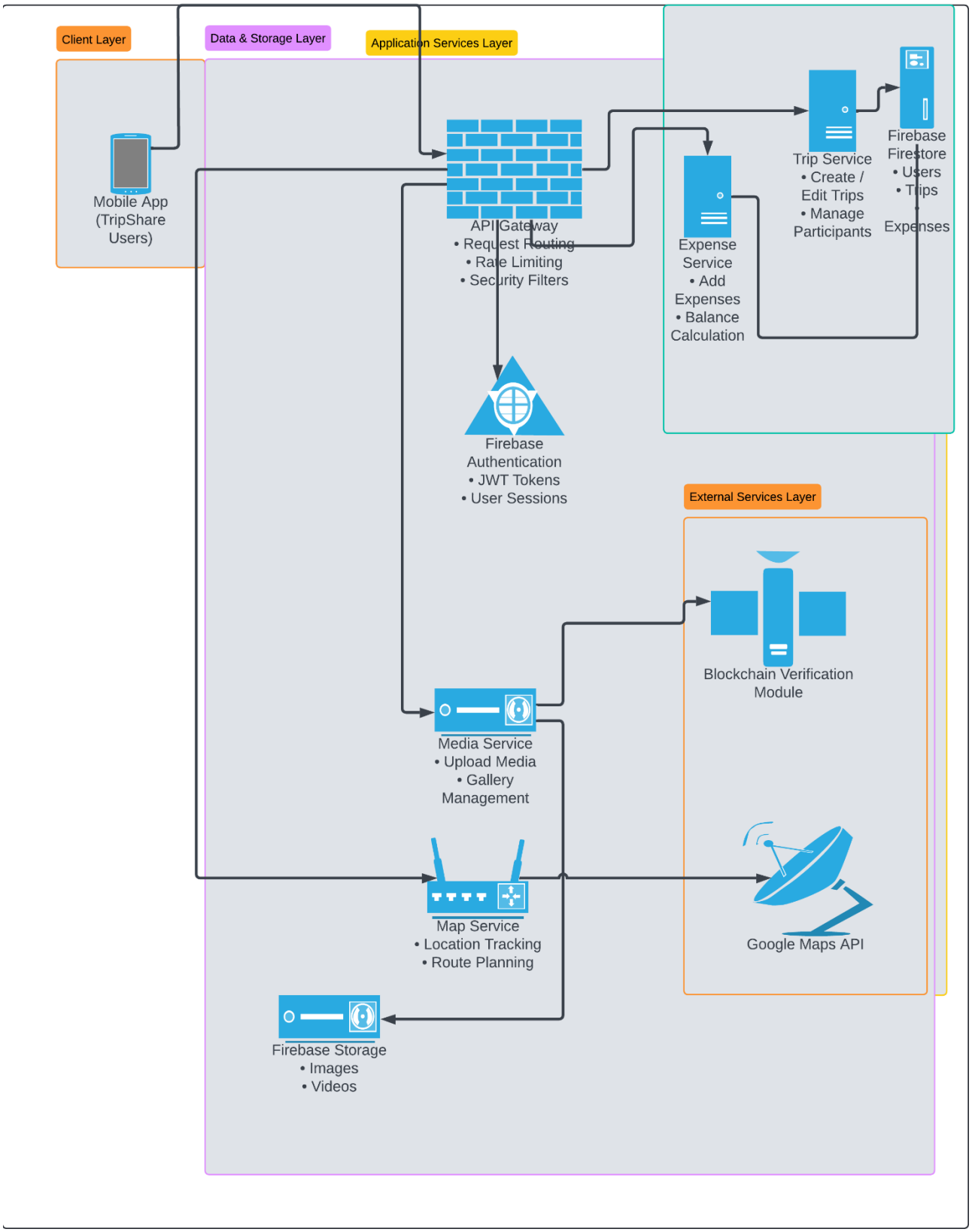

Figure 3: Layered Component Architecture of the Whisper Platform

6.4 Activity Diagram

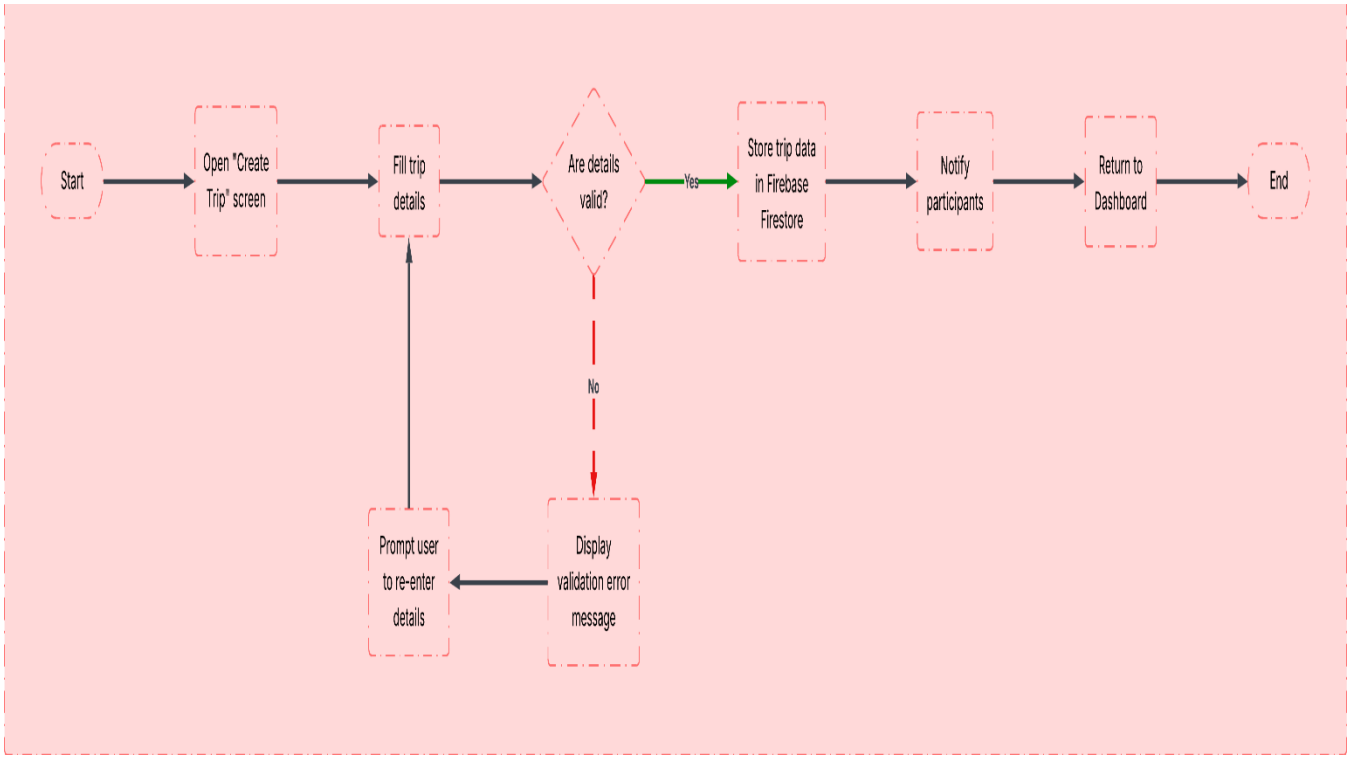

Figure 4: Activity Diagram listing the Expert Qualification Review and Confirmation Workflow

6.5 Sequence Diagram

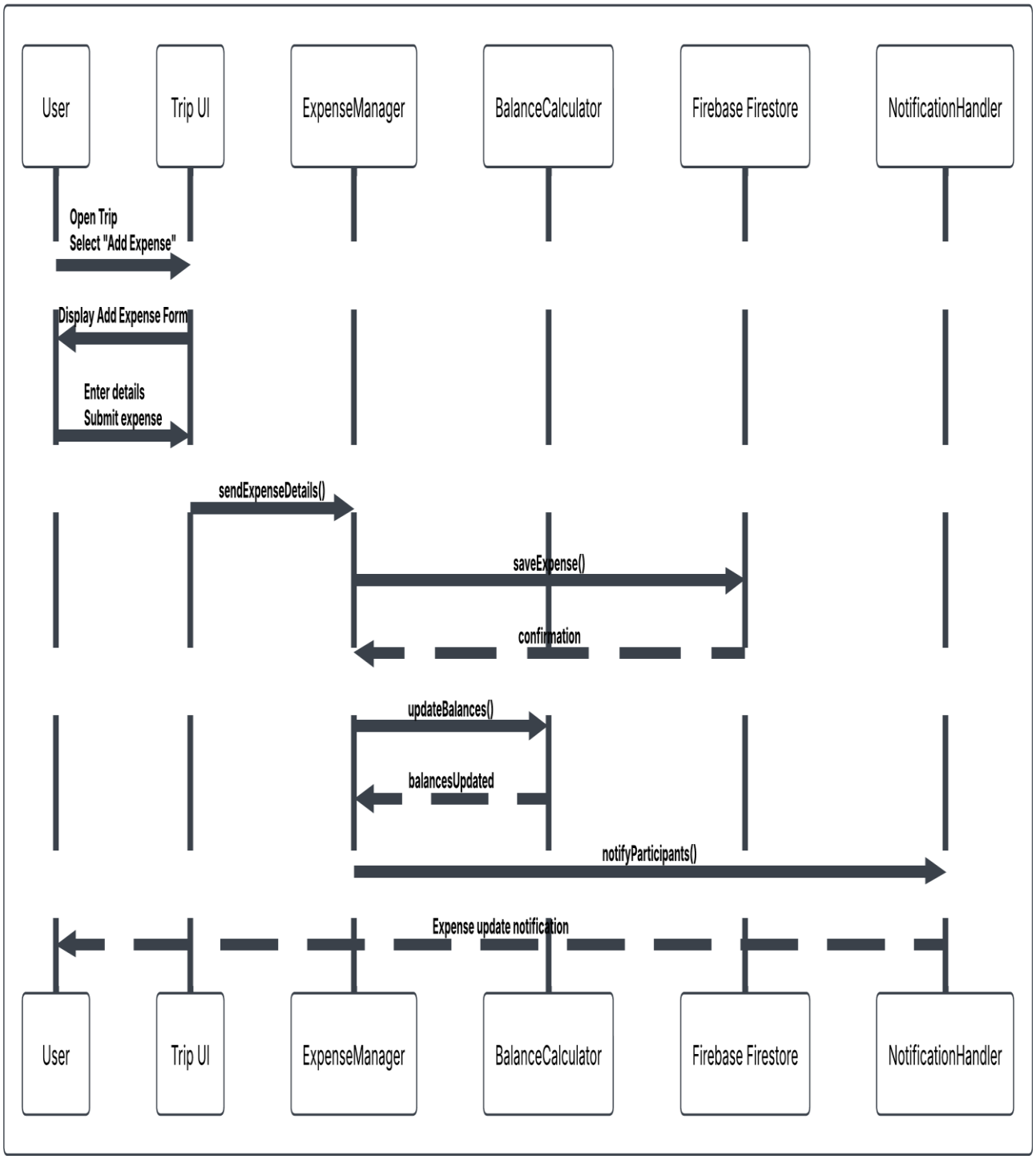

Figure 5: Sequence Diagram for Anonymous Question Submission and Processing Flow

6.6 Component Diagram

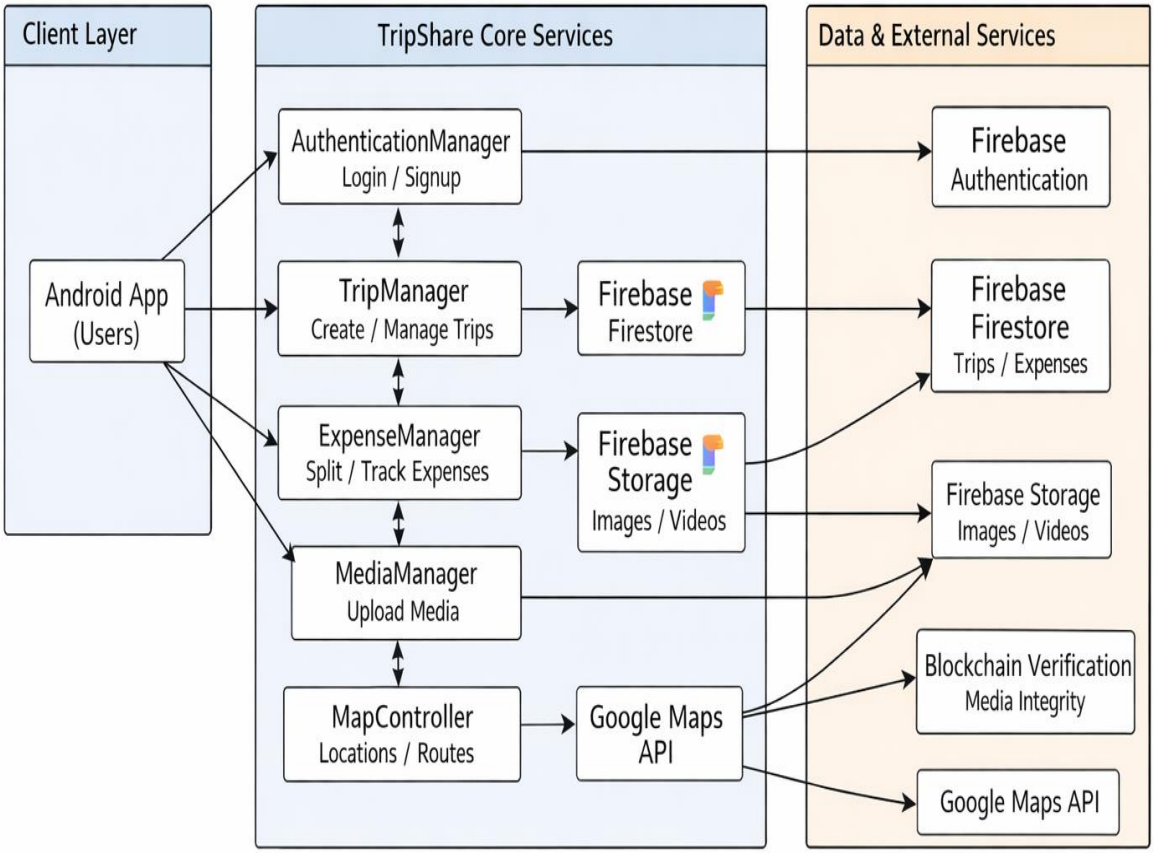

Figure 6: Component Diagram showing the Software Modules and their Dependences

6.7 State Machine Diagram

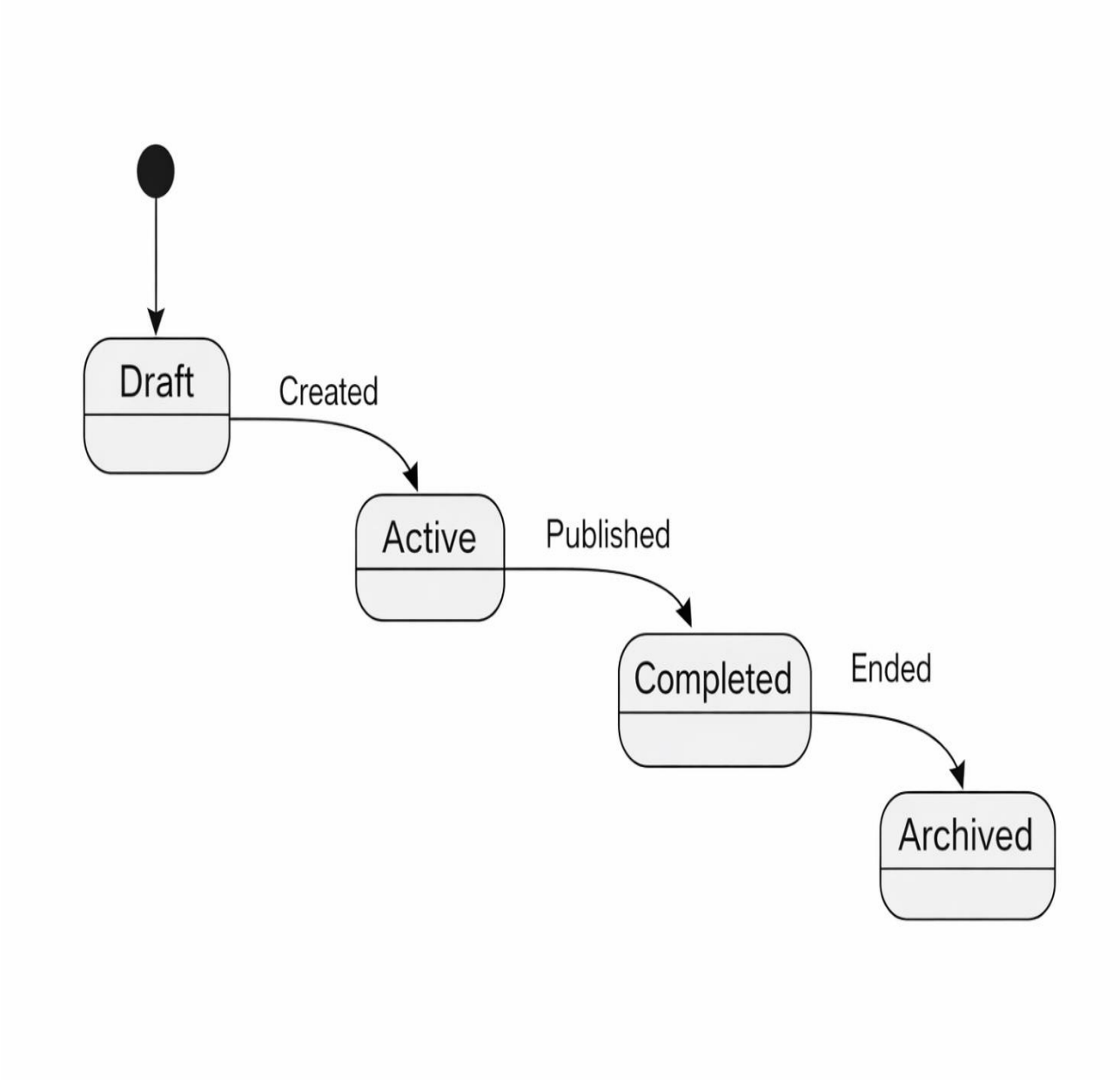

Figure 7: State Machine Diagram for the Consultation Question Lifecycle

6.8 Class Diagram

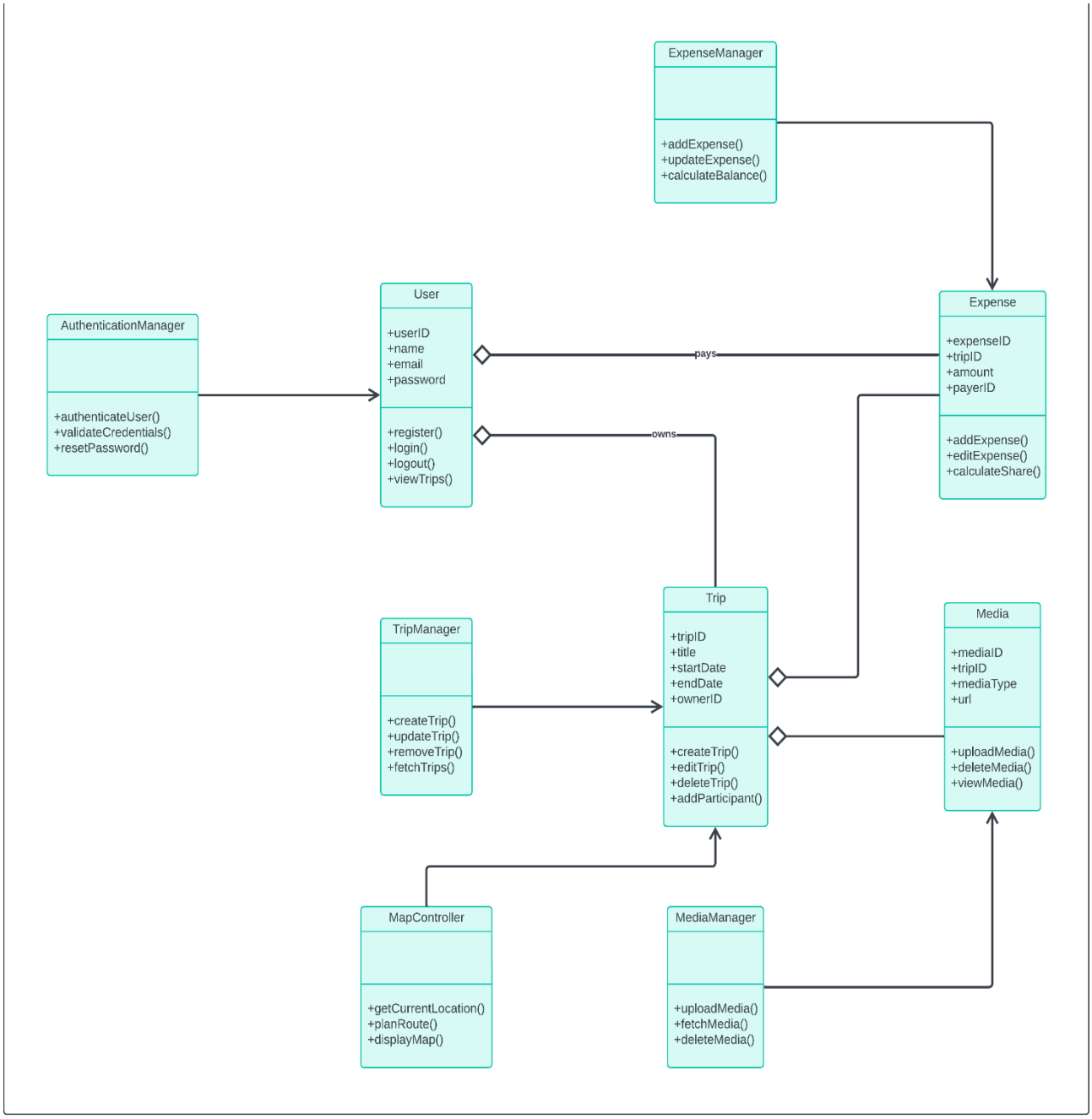

Figure 8: Class Diagram of Core System Entities

6.9 Data Flow Diagram

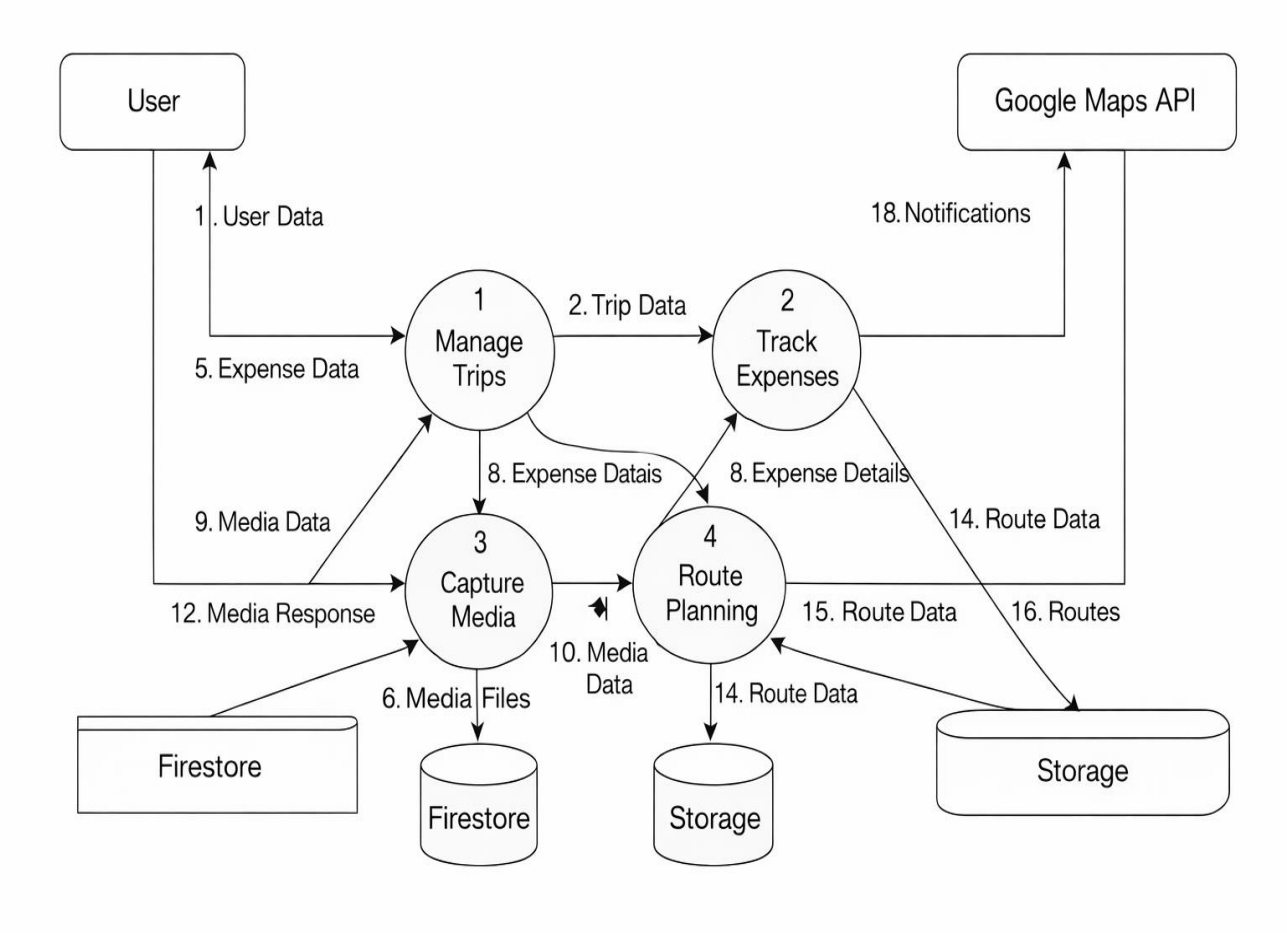

Figure 9: Data Flow Diagram (DFD) show Data Flow between the System and Outside Objects

6.10 Database Diagram

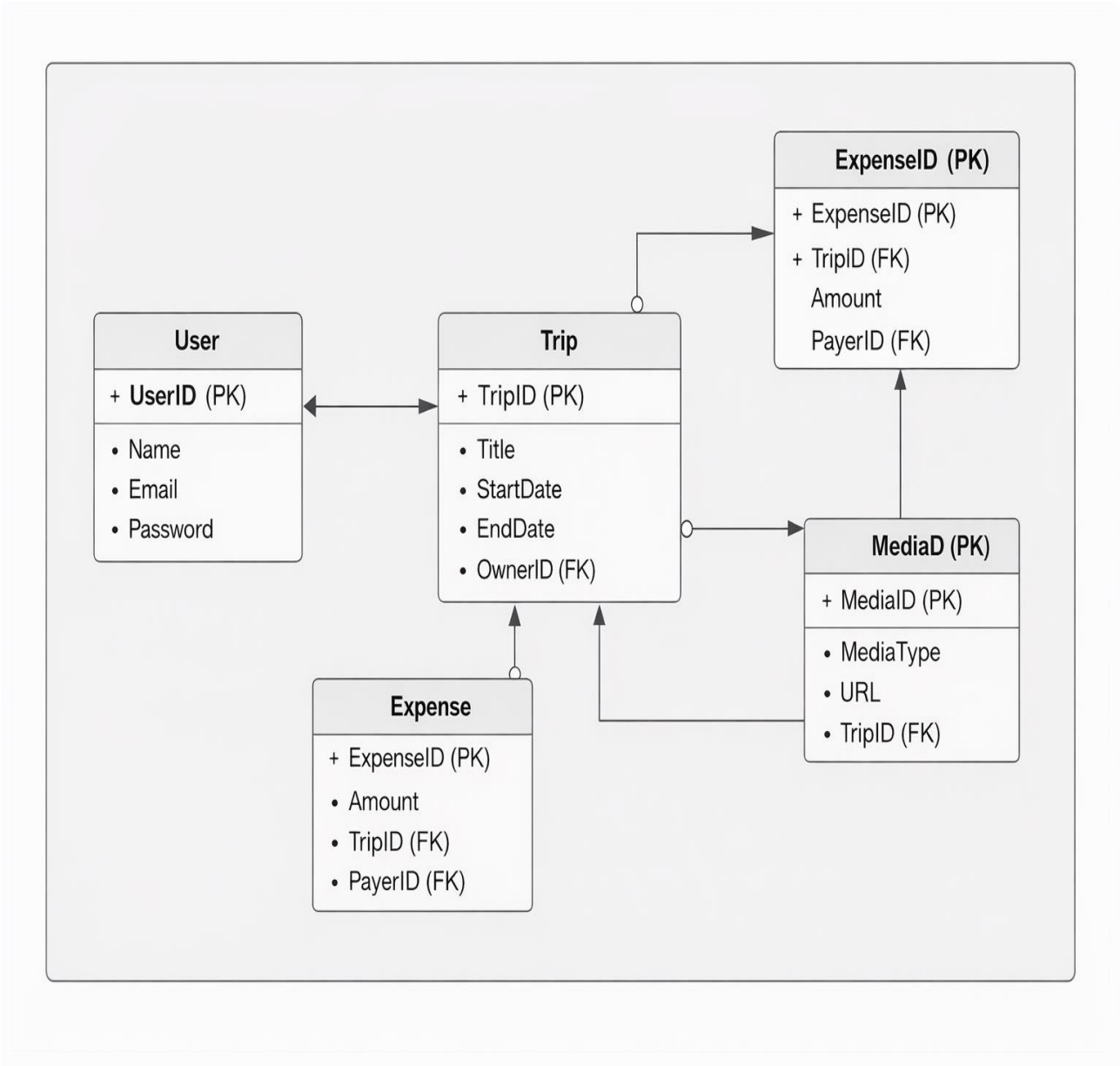

Figure 10: Database Diagram listing the Physical Tables, Qualities, and External Keys in MySQL

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